

Overview of the BRICS Geopolitics Simulator

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Abstract

The BRICS Geopolitics Simulator is a computational framework designed to model geopolitics as an evolving, data-driven system rather than a collection of isolated policy decisions. By integrating news signals, macroeconomic data, network representations, game-theoretic interactions, and scenario-based risk modeling, the project aims to make complex geopolitical dynamics more interpretable and experimentally tractable. While the simulator formalizes high-level strategic interactions among states, it is motivated by the recognition that geopolitical decisions do not remain confined to elite policy circles. Growing up in South Africa, within a country deeply embedded in the BRICS grouping, I have directly observed how global alignments, trade policies, and financial shifts propagate downward into everyday economic realities. This project therefore combines formal modeling with an awareness of lived experience, emphasizing that geopolitics is simultaneously structural, strategic, and social.

1 Introduction

The core motivation of the BRICS Geopolitics Simulator is the idea that geopolitics can be treated as a computable system. States interact repeatedly, respond to signals, adapt strategies, and generate emergent outcomes over time. Traditional qualitative analysis captures rich contextual detail, but it often struggles to systematically explore counterfactuals or stress-test alternative futures. By contrast, computational models allow researchers to formalize assumptions, run scenarios, and observe system-level behavior under controlled perturbations.

This perspective is informed by computational and agent-based approaches in political science, which treat macro-level outcomes as emerging from structured micro-level interactions among actors¹. The BRICS grouping—Brazil, Russia, India, China, and South Africa—provides a particularly relevant case, as it combines heterogeneous political systems, economic structures, and regional interests within a loose but influential alignment.

2 Personal Motivation and Perspective

Although the simulator operates at the level of states and systems, it is not motivated solely by abstract theory. I grew up in South Africa, where BRICS is not just an acronym discussed in policy papers, but a tangible part of national economic discourse. Exchange rate volatility, shifts

¹For a foundational discussion of agent-based and computational modeling in political science, see [1].

in trade orientation, energy agreements, and global financial conditions linked to BRICS dynamics have visible effects on employment, prices, infrastructure, and public debate.

This lived experience informs the project’s philosophy: geopolitical decisions are not only made in high-level diplomatic meetings, but also “trickle down” through economic and social channels. The simulator is therefore designed not merely to describe elite strategy, but to help reason about downstream consequences and systemic risk propagation.

3 System Architecture

At a high level, the simulator follows a modular pipeline:

News → Signals → Knowledge Graph → Game Theory → Risk Model → Scenario Engine →
Dashboard

Each stage transforms information into a more structured representation. News and macroeconomic data are converted into numerical signals; signals update a geopolitical knowledge graph; strategic interactions are simulated through game-theoretic models; and systemic risk is evaluated under baseline and perturbed scenarios.

4 Data Sources

The simulator integrates several broad classes of data:

- World Bank and similar international macroeconomic indicators.
- Google News RSS and related media feeds for real-time signal extraction.
- Chokepoint exposure and trade-dependency datasets.
- Global risk and stress indicators.

Textual data is processed using established techniques from textual analysis, including keyword frequencies and sentiment-weighted measures ².

5 Core Components

5.1 News Signal Extraction

Unstructured text is transformed into structured signals representing intensity, sentiment, and topic relevance. These signals act as short-term information shocks that influence strategic behavior.

²Domain-specific textual analysis is widely used in economics and finance; see [2].

5.2 Geopolitical Knowledge Graph

The geopolitical system is represented as a weighted graph, where nodes correspond to states and edges encode alignment, pressure, or coordination. Graph-based representations provide a compact way to model indirect effects and network externalities ³.

5.3 Game-Theory Engine

Strategic interactions are modeled using repeated and adaptive games. Rather than assuming perfectly rational actors, strategies adjust over time in response to observed payoffs and signals.

5.4 Scenario Risk Engine

The risk engine aggregates shocks and propagates them through the system to estimate systemic exposure. This allows for stress-testing under adverse or hypothetical scenarios.

5.5 Interactive Dashboard

Results are presented through an interactive dashboard, enabling users to explore scenario outcomes, compare trajectories, and visualize network dynamics.

6 Mathematical Models (High-Level)

The simulator relies on three core modeling ideas. These models are deliberately kept interpretable and stylized: they are not intended as literal predictors of policy outcomes, but as structured mechanisms for reasoning about how strategic pressure, coordination, and escalation can evolve over time within the BRICS context.

6.1 Adaptive Tariff Game

The Adaptive Tariff Game models trade policy as a repeated strategic interaction. Each state i selects a tariff intensity $a_i \in [0, 1]$, where higher values represent more protectionist or retaliatory trade stances. The payoff to a state depends on its own tariff level, the tariff levels of its trading partners, and domestic economic costs:

$$\pi_i = \alpha_i a_i - \beta_i a_i \sum_{j \neq i} a_j - \gamma_i a_i^2. \quad (1)$$

In applied terms, the first term captures short-run political or fiscal gains from tariffs, the second captures retaliation and trade friction, and the third reflects internal economic costs. Rather than

³For a general treatment of network representations and dynamics, see [3].

assuming instantaneous optimization, the simulator uses a smooth adaptive update:

$$a_i^{(t+1)} = \frac{1}{1 + \exp(-\lambda \pi_i^{(t)})}. \quad (2)$$

This functional form reflects bounded rationality: states gradually adjust policy in response to observed outcomes. In the BRICS context, this mirrors how trade policy often evolves incrementally through negotiation, retaliation, and learning, rather than via one-off optimal decisions.

6.2 Energy & Currency Coordination Game

Energy pricing and currency alignment are modeled as a coalition coordination problem. Let $C \subseteq N$ denote a coalition of states coordinating on pricing, settlement mechanisms, or reserve usage. Coalition utility is defined as

$$U(C) = \sum_{i \in C} u_i - \delta |C|^2, \quad (3)$$

where u_i captures state-specific gains from coordination (e.g. reduced transaction costs or increased bargaining power), and $\delta |C|^2$ represents increasing coordination and political costs.

In application, this structure reflects the reality that while deeper BRICS coordination can increase collective leverage, it also introduces frictions: heterogeneous interests, institutional complexity, and political risk. The model allows coalitions to form, expand, or fragment depending on whether marginal benefits exceed marginal coordination costs.

6.3 Richardson-style Tariff Arms Race

Escalatory dynamics are modeled using a continuous-time system inspired by Richardson's arms race equations. Let $x(t)$ and $y(t)$ denote tariff or trade-barrier intensities for two blocs:

$$\frac{dx}{dt} = ay - bx + r, \quad (4)$$

$$\frac{dy}{dt} = cx - dy + s. \quad (5)$$

Here, the cross terms ay and cx represent reactive escalation, while bx and dy represent internal economic or political restraint. The constants r and s capture baseline grievances or external pressures.

In practical terms, this model helps distinguish regimes of stability from runaway escalation. When restraint dominates reaction, the system converges to a stable equilibrium; when reaction dominates, tariffs spiral upward. This is particularly useful for exploring stress scenarios in which trade conflicts intensify across multiple BRICS-related supply chains⁴.

⁴Arms race dynamics originate in Richardson's work; see [4].

7 Use Cases

The primary use cases of the simulator fall into three categories. First, scenario analysis allows users to explore how alternative assumptions—such as increased energy coordination or intensified trade conflict—shape system-wide outcomes. Second, stress-testing evaluates how shocks propagate through interconnected trade, currency, and geopolitical networks. Third, visualization tools enable users to inspect alignment and fragmentation within BRICS over time.

These use cases are particularly valuable for comparative “what-if” analysis, where the goal is not prediction, but structured exploration of plausible futures under clearly stated assumptions.

8 Limitations and Interpretive Boundaries

The simulator necessarily simplifies political behavior. Domestic politics, institutional inertia, and informal power dynamics are abstracted into compact parameters. Results depend on data quality, signal extraction choices, and parameterization. For these reasons, outputs should be interpreted as exploratory patterns rather than forecasts.

Importantly, the simulator is designed to support reasoning, not authority. It makes assumptions explicit and exposes how conclusions change when those assumptions are altered.

9 Conclusion

The BRICS Geopolitics Simulator offers a structured computational framework for reasoning about strategic interaction, coordination, and escalation in a complex multipolar environment. While grounded in formal mathematical models, the project is motivated by lived experience—including the visible ways BRICS-related decisions in South Africa propagate into everyday economic and social conditions. By combining formal structure with contextual awareness, the simulator emphasizes that geopolitics is not only a system of elite strategy, but a process whose effects ultimately reach far beyond high-level decision-making.

References

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